JOSEPH ESTEBAN POLANIA TRIVIÑO

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About me

Mechanical and Industrial Engineer with experience in manufacturing processes, process optimization, high-precision CAD design, quality control, and automation. Experienced in production environments, statistical process control, technical documentation, and multidisciplinary engineering projects. Currently pursuing a Master's degree in Space Technology and Business at the University of Luxembourg, with additional hands-on experience in robotic and electromechanical systems. Strong analytical and problem-solving skills, with a Lean Six Sigma mindset and an interest in industrial automation, digital manufacturing, and Industry 4.0.

Education & Training

MASTER IN SPACE TECHNOLOGIES AND BUSINESS | University of Luxembourg | 12/09/2025 - Current | Luxembourg, Luxembourg

MECHANICAL ENGINEER | University of La Sabana | 25/01/2016 - 22/11/2022 | Chia, Colombia

INDUSTRIAL ENGINEER | University of La Sabana | 25/01/2016 - 22/11/2022 | Chia, Colombia

Work experience

RESEARCH STUDENT ASSISTANT | University of Luxembourg — SpaceR Research Group | 01/03/2026 - Current | Luxembourg, Luxembourg

- Design and implement control architectures for robotic manipulation and autonomous grasping systems.
- Develop and integrate hardware and software components including motors, tactile sensors, controllers, and robotic end-effectors.
- Perform Finite Element Analysis (FEA) to evaluate the deformation, stress, and mechanical performance of compliant mechanisms for robotic end-effectors
- Develop modular and maintainable system architectures with a focus on reliability and repeatability.

CAD DESIGN ENGINEER (IMAGING ENGINEER PSI) | Zimmer Biomet | 01/12/2022 - 27/08/2025 | Bogotá, Colombia

- Developed and validated high-precision 3D CAD models from medical imaging data for engineering and manufacturing applications
- Designed patient-specific surgical guides using controlled CAD-based engineering workflows, ensuring dimensional accuracy and repeatability
- Supported process standardization and technical documentation for engineering workflows
- Delivered technical onboarding and process training to new engineering personnel

PLASTIC TECHNOLOGY INTERN | Organización Corona | 08/02/2022 - 07/08/2022 | Funza, Colombia

- Supported the technical and financial planning of a plastic raw-material recovery project within a manufacturing plant
- Developed a Statistical Process Control (SPC) approach for plastic injection machines to support process monitoring, quality improvement, and identification of process variability
- Analyzed manufacturing processes and production data to identify opportunities for process optimization and waste reduction
- Collaborated with manufacturing and engineering teams to translate process requirements into practical improvement initiatives

INNOVATION LAB HOST: 3D PRINTING SYSTEM | University of La Sabana | 17/01/2021 - 10/12/2021 | Chia, Colombia

3D Printing Service Management

Skills

Statistical Quality Control | Lean Six Sigma Methodology | SolidWorks | Autodesk Inventor | AutoCAD | LabView | TIA Portal | ROS2 | Matlab/Simulink | Python | Ansys (FEM) | Aditive Manufacturing

Projects

Universal Gripper Control System

Developed a control architecture for an autonomous robotic manipulation system integrating motor actuation, force/tactile sensing, feedback control, and ROS 2.

Continuous Cough and Snoring Monitoring Instrument

Development of a prototype embedded system for continuous acoustic monitoring and signal analysis.

Pet Food Production Facility Design

Designed a production facility and workflow with a focus on manufacturing efficiency, process flow, layout optimization, and resource utilization.

Leadership and Awards

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- President — Institute of Industrial and Systems Engineers Student Chapter (2020–2021)
- Vice President — American Society of Mechanical Engineers Student Chapter (2020–2021)
- Winner — "Dreamland" Business Challenge, University of La Sabana Center for Entrepreneurship and Innovation (2020)